In [7]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(r"C:\Users\praka\Dataset .csv")
```

In [8]:
```
df.head()
```

Out[8]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...

5 rows × 21 columns

In [9]:
```
df.tail()
```

Out[9]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality
9546	5915730	Naml\ Gurme	208	××stanbul	Kemanke×× Karamustafa Pa××a Mahallesi, R\ht\m ...	Karak×_y
9547	5908749	Ceviz A××ac\	208	××stanbul	Ko××uyolu Mahallesi, Muhittin ××st×_nda×× Cadd...	Ko××uyolu
9548	5915807	Huqqa	208	××stanbul	Kuru×_e××me Mahallesi, Muallim Naci Caddesi, N...	Kuru×_e××me
9549	5916112	A×××k Kahve	208	××stanbul	Kuru×_e××me Mahallesi, Muallim Naci Caddesi, N...	Kuru×_e××me
9550	5927402	Walter's Coffee Roastery	208	××stanbul	Cafea××a Mahallesi, Bademalt\ Sokak, No 21/B, ...	Moda

5 rows × 21 columns

In [10]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Restaurant ID         9551 non-null   int64
 1   Restaurant Name       9551 non-null   object
 2   Country Code          9551 non-null   int64
 3   City                  9551 non-null   object
 4   Address               9551 non-null   object
 5   Locality              9551 non-null   object
 6   Locality Verbose      9551 non-null   object
 7   Longitude             9551 non-null   float64
 8   Latitude              9551 non-null   float64
 9   Cuisines              9542 non-null   object
 10  Average Cost for two  9551 non-null   int64
 11  Currency              9551 non-null   object
 12  Has Table booking     9551 non-null   object
 13  Has Online delivery   9551 non-null   object
 14  Is delivering now     9551 non-null   object
 15  Switch to order menu  9551 non-null   object
 16  Price range           9551 non-null   int64
 17  Aggregate rating      9551 non-null   float64
 18  Rating color          9551 non-null   object
 19  Rating text           9551 non-null   object
 20  Votes                 9551 non-null   int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

In [11]:
```
df.describe()
```

Out[11]:

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	
count	9.551000e+03	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9
mean	9.051128e+06	18.365616	64.126574	25.854381	1199.210763	1.804837	
std	8.791521e+06	56.750546	41.467058	11.007935	16121.183073	0.905609	
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	
25%	3.019625e+05	1.000000	77.081343	28.478713	250.000000	1.000000	
50%	6.004089e+06	1.000000	77.191964	28.570469	400.000000	2.000000	
75%	1.835229e+07	1.000000	77.282006	28.642758	700.000000	2.000000	
max	1.850065e+07	216.000000	174.832089	55.976980	800000.000000	4.000000	

In [12]:
```
df.shape
```

Out[12]:
```
(9551, 21)
```

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
Restaurant ID           0
Restaurant Name         0
Country Code            0
City                    0
Address                 0
Locality                0
Locality Verbose        0
Longitude               0
Latitude                0
Cuisines                9
Average Cost for two    0
Currency                0
Has Table booking       0
Has Online delivery     0
Is delivering now       0
Switch to order menu    0
Price range             0
Aggregate rating        0
Rating color            0
Rating text             0
Votes                   0
dtype: int64
```

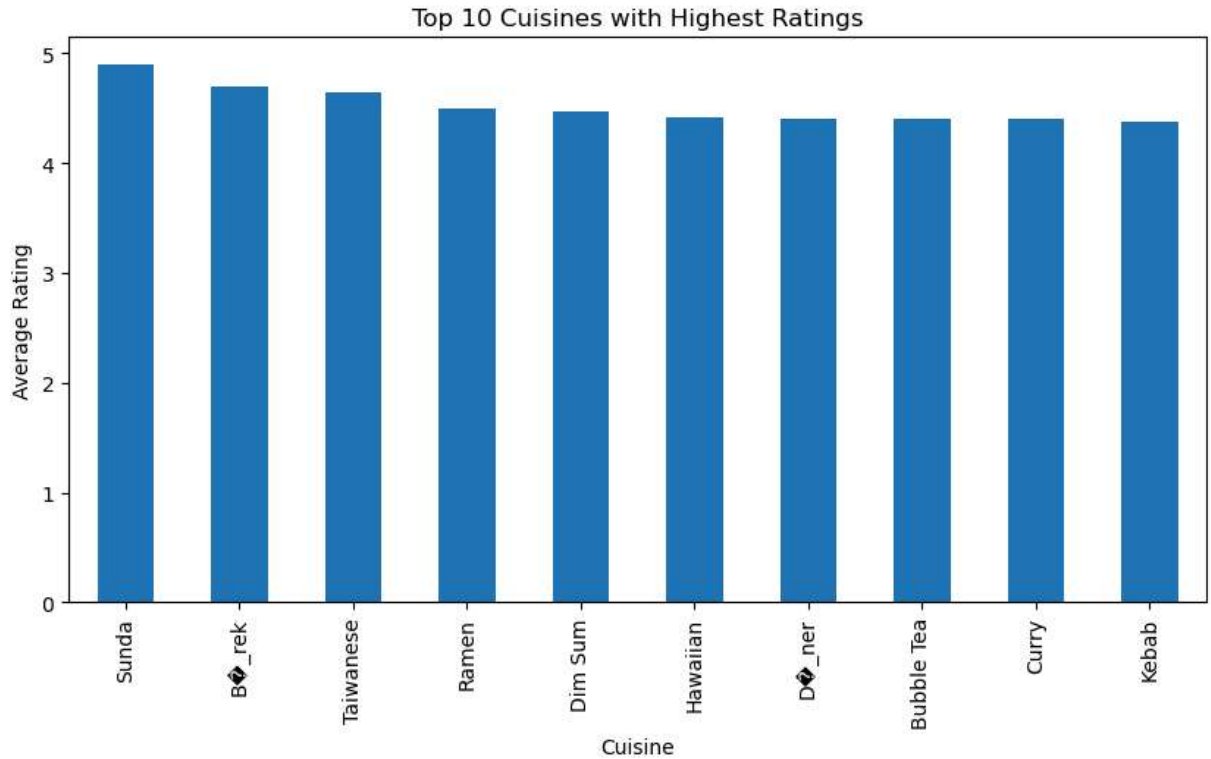
In [14]:
```
df.duplicated()
```

Out[14]:
```
0       False
1       False
2       False
3       False
4       False
        ...
9546    False
9547    False
9548    False
9549    False
9550    False
Length: 9551, dtype: bool
```

In [15]:
```
df['Cuisines'] = df['Cuisines'].str.split(', ')
df_exploded = df.explode('Cuisines')
```

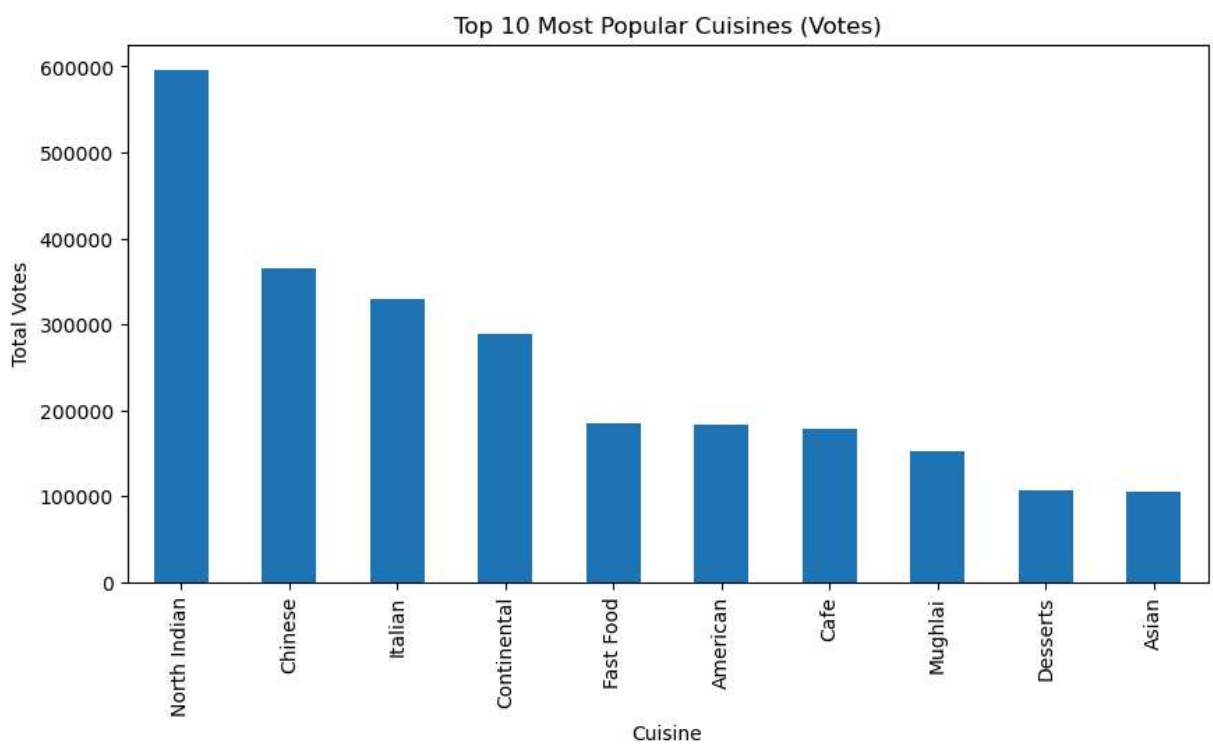
In [16]:
```
cuisine_rating = df_exploded.groupby('Cuisines')['Aggregate rating'].mean().sort_va

cuisine_rating.head(10).plot(kind='bar', figsize=(10,5))
plt.title("Top 10 Cuisines with Highest Ratings")
plt.xlabel("Cuisine")
plt.ylabel("Average Rating")
plt.show()
```



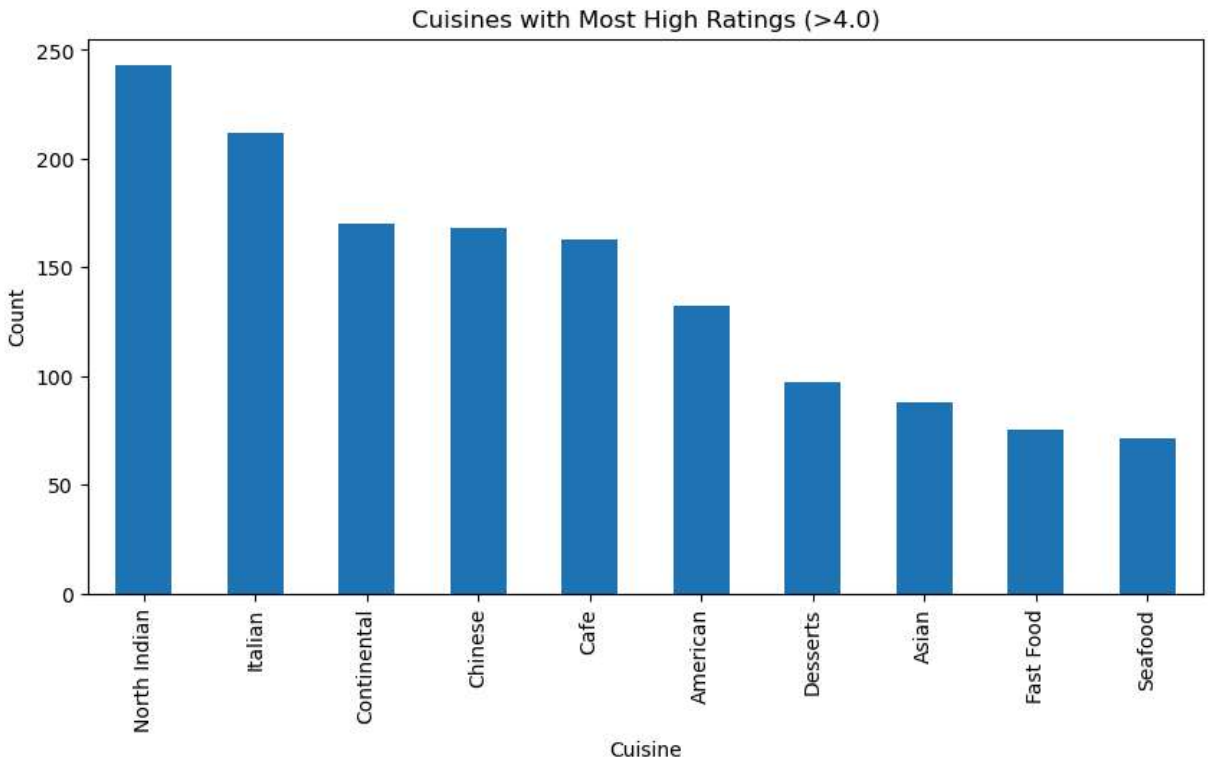
In [17]:
```
cuisine_votes = df_exploded.groupby('Cuisines')['Votes'].sum().sort_values(ascendin

cuisine_votes.head(10).plot(kind='bar', figsize=(10,5))
plt.title("Top 10 Most Popular Cuisines (Votes)")
plt.xlabel("Cuisine")
plt.ylabel("Total Votes")
plt.show()
```



In [18]:
```
high_rated = df_exploded[df_exploded['Aggregate rating'] > 4.0]

high_rated['Cuisines'].value_counts().head(10).plot(kind='bar', figsize=(10,5))
plt.title("Cuisines with Most High Ratings (>4.0)")
plt.xlabel("Cuisine")
plt.ylabel("Count")
plt.show()
```



In [ ]: